

Perspectives on Insurance and Climate Risk

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Abstract

Property insurance is central to managing rising damage from climate change. Yet recent disruptions in insurance markets have raised concerns about their long-term viability, with significant implications for households, real-estate markets, and the distribution of climate risk in the economy. While property insurance markets share common features with other well-studied markets, several central features make it distinct: uncertain and correlated risks, close ties to housing and mortgages, and the policy environment. This review highlights emerging evidence and proposes research directions to advance the homeowners insurance agenda by integrating insurance into the broader climate finance, household finance, and macro-finance literature.

1. Introduction

Scientific projections suggest that losses from property damage will continue to escalate as climate change brings more extreme temperatures and increases both the frequency and severity of natural disasters ([IPCC \(2021\)](#)).¹ Property insurance is a key tool for managing these rising risks. Given the importance of insurance in a world with climate change, a fundamental question is: how will rising climate risks affect property insurance markets?

Under the benchmark model of insurance, rising risks present large opportunities for insurance markets. All else equal, rising risks increase household demand for insurance and the welfare benefits from risk sharing—services we consider to be essential functions of financial markets. Yet, disruptions in insurance markets are now frequently front-page news, developments that are difficult to reconcile with the standard model: households are

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¹Swiss Re estimates that climate change contributed to roughly 30% of the recent growth in losses. [Swiss Re, "Fact Sheet", March 2024](#).

facing double-digit premium growth;² record numbers can only obtain policies from state-run insurers of last resort;³ insurance companies themselves are raising concerns about their solvency.⁴ There is also ongoing debate about public intervention, including proposals for a federal reinsurance backstop. These developments reflect a growing public concern that insurance markets may face significant challenges as climate risks intensify.

Yet, despite the public discourse and the central role of property insurance in the economy, the academic literature on this topic remains thin. Insurance is one of the oldest topics in economics, with significant work spanning over five decades. However, while there is a robust literature on other markets, such as health insurance and to some extent life insurance (and annuities), property (homeowners) insurance is not well-studied, either theoretically or empirically. One reason for this gap may be the historical difficulty of obtaining high-quality, granular data. Conceptually, the economics of property insurance has important parallels with other well-studied markets, particularly in the roles of financial frictions and informational asymmetries. However, in our view there are many aspects that make property insurance distinct, including: (i) the uncertain and correlated nature of climate risk, (ii) its linkages with mortgage and housing markets, and (iii) the policy and institutional environment. With recent data advances, the academic literature is well-positioned to shed light on many of these dynamics, creating an extraordinary opportunity for research.

Understanding the economics of property insurance is important for a number of reasons. First, it is a large market: nearly 83% of American households have homeowners insurance, with over \$15 trillion in coverage outstanding (Sastry et al. (2024)). Second, property insurance is a critical tool to preserve home values, the primary source of wealth for most households, against climate-related damage. A large share of homeowners face heightened exposure, with some estimates indicating that more than one-third of U.S. properties are located in high-risk areas.⁵ Property insurance is therefore central for preserving household financial resiliency to rising climate risks. Third, insurance is required to obtain a mortgage, making insurance the backbone of the mortgage market and, in turn, the real-estate market. As a result, disruptions in property insurance can propagate through housing and financial markets, affecting who bears climate risk and the stability of the broader economy.

In this review, we propose multiple lines of inquiry in the homeowners insurance research

²The Insurance Journal, “Florida Property Insurance Premiums Surged 34% Since 2022”, June, 2025.

³The Insurance Journal, “State Residual Markets See Policy Count Double in 5 Years”, October, 2024.

⁴State Farm writes: “We remain deeply concerned about the financial position of State Farm General, as it is difficult to match price to risk in California.” State Farm, Press Release, August 4, 2025.

⁵CoreLogic’s 2020 catastrophe report estimated that 35 million homes, or roughly one-third of the housing stock, are at risk of a natural disaster. Other recent estimates are also similar, see 2025 Realtor.com Housing and Climate Risk Report, September 3, 2025.

agenda that we believe can contribute to the broader understanding of the fundamental challenges in these markets. We outline how we think about the economics of the property insurance markets on the supply side and the demand side, as well as connections between insurance and the broader economy. An important question is how much of the observed patterns in the data can be explained by the existing benchmark models of insurance, and whether we may need to posit new theories that account for other possible deviations from the model’s assumptions. We caution that much of our discussion will, by necessity, be quite *speculative*. The literature on homeowners insurance is just emerging, and our primary focus is to highlight opportunities and priorities for future work on this topic. That said, we will highlight existing scholarship on homeowners insurance as much as possible, and where relevant we also draw comparisons to other insurance and credit markets.

Roadmap: Our review of the literature is organized as follows. We begin in Section 2 with the benchmark model of insurance, to set ideas about the risk sharing benefits of insurance markets. We then consider deviations from the benchmark model on both the supply side (Section 3) and the demand side (Section 4), focusing on how climate change may interact with policy and economic frictions to influence insurance market outcomes. In Section 5, we consider the role that insurance plays in the broader economy (e.g., through its impact on mortgage/housing markets). Finally, in Section 6, we discuss additional topics and conclude with an overview of recent data advances.

Scope and terminology: We use the phrase “property insurance” to primarily refer to homeowners insurance, and occasionally flood insurance.⁶ Homeowners insurance are contracts where insurers pay out claims if a loss event (disaster) materializes in exchange of up-front premiums paid by households, typically on a monthly basis. Contracts are short-dated (typically 1-2 years). Coverage limits determine the maximum dollar amount that an insurer will pay out in claims, and the deductible on the contract determines the amount the household has to pay before insurers begin paying out claims.⁷ We focus primarily on the United States, where homeowners insurance is mostly provided by private companies (rather than the government); property insurers are separate entities from banks; and flood risk is carved out from the standard contract.⁸ We limit our discussion of climate change to its impact on insurers’ liabilities. We leave questions related to insurers’ asset allocation, other property and casualty markets, international comparisons, and the impact of insurers’

⁶Other property and casualty insurance products, such as auto insurance, will not be covered.

⁷Insurance contracts can vary in other dimensions. For example, some contracts cover a restricted set of perils (e.g., fire-only policies), and others do not cover property depreciation (e.g., actual cash value policies).

⁸Property insurance markets follow different structures in different countries. For example, in Norway it is provided by the government (Espegren et al. (2025)), and in the UK, banks provide both mortgage and insurance products. In the U.S., flood insurance is provided primarily by the government.

underwriting on carbon emissions for future work.

2. Background and Conceptual Framework

To organize our thoughts, we begin with the benchmark model of insurance ([Arrow \(1963\)](#), [Pratt \(1964\)](#), [Mossin \(1968\)](#)). In this one-period model, the household with concave utility function U and initial wealth W , faces some idiosyncratic risk of a wealth loss, represented by $\tilde{L}$. The household utility function can be represented as $U(W - \tilde{L})$.

Demand. The household considers whether to buy insurance to fully cover its loss exposure for a premium p . The household’s willingness-to-pay (π) comes from its certainty equivalent for fully insuring this risk, which equates the household’s expected utility with the risk, and household’s utility after paying the premium. The household’s willingness-to-pay for full insurance satisfies:

$$(1) \quad \pi \approx \mathbb{E}[\tilde{L}] - \frac{1}{2} \cdot \frac{U''(W)}{U'(W)} \mathbb{E}[\tilde{L}^2].$$

Equation 1 shows that household’s willingness-to-pay is equal to her expected loss plus her utility cost of bearing idiosyncratic risk, which is a function of the household’s risk aversion and the variance of the risk.⁹ In this framework, households buy full insurance at actuarially fair prices, and their willingness-to-pay for insurance is increasing in their risk exposure as well as risk aversion.¹⁰

Supply. Insurance is provided by a fully diversified insurer in a frictionless, perfectly competitive market. Because risk is assumed to be idiosyncratic, full diversification makes the insurer effectively risk neutral over idiosyncratic risks ([Samuelson \(1963\)](#), [Ross \(1976\)](#), [Diamond \(1984\)](#)).¹¹ Full diversification also limits the need for the insurer to bear any other hedging costs, since under actuarially fair pricing the probability of insurer insolvency diminishes with diversification ([Arrow \(1963\)](#)). Thus, the diversified insurer optimally sets premiums equal to the expected loss.

Welfare and Allocative Efficiency. This simple model yields rich predictions about

⁹This is derived with a Taylor expansion around the level of wealth W . The utility cost is often referred to in the insurance economics literature as an (idiosyncratic) risk premium ([Einav and Finkelstein \(2011\)](#)).

¹⁰Consistent with the model’s predictions, a large literature uses revealed-preference approaches using data on contract choices and claims to uncover households’ structural parameters, including willingness-to-pay, risk aversion, and risk exposures ([Cicchetti and Dubin \(1994\)](#), [Cohen and Einav \(2007\)](#), [Sydnor \(2010\)](#)).

¹¹Specifically, [Diamond \(1984\)](#) shows that under decreasing absolute risk aversion preferences, the certainty equivalent per gamble increases when gambles are pooled, meaning the insurer would require a lower risk premium per policy when holding many independent policies than when holding one. Effective risk aversion declines as wealth increases or as other imperfectly correlated risks are added to a diversified portfolio.

insurance demand and supply, the welfare benefits of risk-sharing, and how these forces evolve as climate risk rises. All else equal, rising risk (e.g., changes in the mean or variance of the loss distribution) raises households' willingness to pay for insurance, either because their climate exposure increases or because they become more risk-averse. On the supply side, rising expected losses lead insurers to charge higher premiums. Crucially, the risk averse household is willing to pay an insurance premium that exceeds the insurer's marginal cost. Therefore, insurance purchased at actuarially fair rates increases household utility and leaves the insurer indifferent, creating gains from trade.

However, the standard model reflects insurance markets in an idealized competitive setting with idiosyncratic risks and without frictions or policy distortions. A number of fundamental assumptions of the standard model may not hold in practice, which can influence how insurance markets respond to rising climate risk. For example, insurance markets are heavily regulated, and growing evidence shows that insurers operate under financial constraints, which may worsen further if risks become more aggregate in nature due to climate change. On the demand side, most households buy their homes using a mortgage, making mortgage market considerations an important factor in insurance demand. These deviations may change the economics of both the supply-side and the demand-side of the insurance market, raising questions about what risk sharing will look like in a world with rising climate risks.

It is also important to consider the role of property insurance in the broader macroeconomy. In the absence of well-functioning insurance markets, risks may spill over from the insurance sector to other parts of the economy, reshaping ultimately who bears climate risk. Furthermore, insurance prices determine how various agents in the economy internalize the risks of living in different locations; mis-pricing can therefore alter the geography of economic activity. We examine these deviations and their real effects throughout the rest of this review.

3. The Supply of Insurance

In this section, we consider how rising climate risk may impact insurance supply. In the frictionless benchmark model, insurers would respond by only adjusting prices, in line with shifts in their expectations of losses. In practice, there is significant evidence that insurers also adjust on other margins: by rationing and exiting high risk areas. There is also evidence that prices are often tied to actuarial levels. We consider deviations from the benchmark model that help explain the price and non-price adjustments made by insurers: uncertainty and correlated risks (Section 3.1), financial frictions (Section 3.2), policy and regulatory frictions (Section 3.3), and market incompleteness (Section 3.4). We survey the literature,

where it is available and also provide speculative thoughts about how these deviations might relate to climate change and influence the supply choices of property insurers.

3.1. Uncertainty and Correlated Risks

The benchmark model assumes that risks are independent and therefore diversifiable, allowing insurers to invoke the law of large numbers to pool idiosyncratic risks across a broad portfolio (Samuelson (1963); Diamond (1984)). Natural disasters, however, are rare events characterized by substantial spatial and temporal dependence, which makes reliable parameter estimation and loss modeling inherently difficult (Weitzman (2009); Kousky and Cooke (2012); Xue et al. (2015)). Parameter uncertainty can generate fat-tailed loss distributions, introducing serious measurement and estimation challenges.¹²

Climate change likely creates a number of sources of uncertainty for insurers, including parameter uncertainty as well as structural uncertainty, in which the underlying distribution itself may be unknown (Barnett et al. (2020)). To navigate these challenges, insurers increasingly rely on forward-looking catastrophe models to assess how climate change may reshape loss distributions over time, including the possibility of nonlinear dynamics and tipping points. These models—developed either in-house or by third-party vendors—often diverge in their methodologies, data inputs, and parameter calibrations, assumptions that are difficult to test or validate empirically (Boomhower et al. (2023); Brunetti et al. (2022); National Association of Insurance Commissioners (2025)). A further complication is that catastrophe models are typically trained on aggregate data and designed to capture longer-term risk patterns, whereas insurers must set annual, property-level prices (Stanton et al. (2025); President’s Council of Advisors on Science and Technology (2023)). Uncertainty around forming risk assessments is likely to be a central challenge, influencing insurance pricing and market outcomes. Insurers may charge uncertainty premia or even ration coverage when such uncertainty interacts with financial frictions.

Compounding these challenges is growing evidence that natural disasters are becoming increasingly correlated across space and time (IPCC (2021)). When risks are correlated, many of the intuitions of the standard model become more nuanced, and depend on what is assumed about the correlation structure of risks.

Standard asset pricing intuition implies that if natural disaster risks are spatially correlated but not correlated with aggregate risk factors, insurance prices would still equal expected loss and will not carry any risk premium. This is because shareholders of the

¹²Even if the true distribution is thin-tailed, Weitzman (2009)’s “dismal theorem” implies that parametric uncertainty can fatten the Bayesian posterior for the mean. Additional feedback mechanisms can similarly produce fat tails (Pindyck (2011)).

insurance company can diversify disaster risks in capital markets. On the other hand, if climate risk starts to become correlated with aggregate risk factors then insurers would also charge an aggregate risk premium above expected losses, as disaster risk can no longer be diversified.¹³ Relatedly, if the financing costs of insurers are correlated with aggregate risk factors, insurers will also charge an aggregate risk premium, regardless of whether disaster losses themselves are correlated with aggregate risk factors (Froot et al. (1999)).¹⁴ A central challenge for the literature is to disentangle how much of the recent increase in insurance premiums reflects rising climate risk versus a growing correlation between climate risk and aggregate economic conditions.

3.2. Financial Frictions

There is significant evidence that insurers face costly external financing constraints. These costs typically spike after large disaster losses, making it difficult for insurers to raise capital exactly when they need to pay claims, thereby driving up insurance prices (Froot et al. (1993), Gron (1994), Winter (1994), Gron (1995), Jaffee and Russell (1997), Gron and Lucas (1998) Froot et al. (1999), Oh et al. (2025)). Evidence from life and annuities markets also show that financial frictions play a key role in pricing (Kojen and Yogo (2015), Ge (2022)). The increasing frequency and severity of disasters are likely to push insurers into financially constrained states more often. One channel could come from climate change worsening the information asymmetries between shareholders and insurers (Holmstrom and Tirole, 1997). This could occur because it may be difficult for shareholders to obtain information on insurers’ geographic exposures and reinsurance contracts. Another channel could manifest from an increase in risk aversion over large catastrophic shocks that threaten insurers’ solvency (Jaffee and Russell, 1997, Pindyck and Wang, 2013).

Importantly, when insurers face financing frictions, rising uncertainty or correlated risks can lead to both higher prices as well as quantity rationing. In this vein, Kojen and Yogo (2022) show that insurers’ financing constraints lead them to ration supply and scale back contract features for insurance contracts that offer protection against aggregate risks. Similarly, Ibragimov et al. (2009) show that the combination of limited liability and fat-tailed disaster risks can also lead to supply-side rationing. There is also emerging evidence that spatially correlated risks lead to higher prices and quantity rationing in the property insurance

¹³For example, Bilal and Känzig (2024)) find that global temperature changes impact macroeconomic variables by changing the distribution of extreme weather events. Giglio et al. (2021) survey the asset pricing literature on whether climate risk is priced in stock returns.

¹⁴As a similar example, consider that biotech firms have a market beta greater than one even though their cash flows are driven by scientific discovery; the high beta comes from the financial constraints and the correlation between their financing conditions and the market factor (Berk and DeMarzo (2019)).

market (Solomon (2025), Keys and Mulder (2024)).

3.3. Policy and Regulatory Frictions

Policymakers have implemented a range of measures to support the functioning of insurance markets. In this section, we highlight four policies that we believe play an outsized role. A distinctive feature of U.S. insurance regulation is that regulation occurs at the state level, creating coordination and political economy challenges that differ from other settings.¹⁵

Financial regulation. There is a long history of financial regulation in insurance aimed at minimizing insurer insolvency risk through minimum capital and loss-reserve requirements and asset-risk oversight. The rationale for financial regulation stems from concerns about moral hazard due the presence of government guarantees (discussed below) and the inability of households to monitor insurers’ risk exposures and long-term solvency. While there is literature examining how capital regulation affects market outcomes in life insurance (Kojen and Yogo (2015), Barbu et al. (2024)), as well as work on the political economy of insurance regulation (Tang (2023), Leverty and Grace (2018), Tenekedjieva (2021)), the literature on property insurance is limited.

A question of central importance is whether risk regulation has kept pace with growing climate risk. Evidence suggests that existing capital requirements may not be sufficiently sensitive to insurers’ risk exposures: several insurers became insolvent despite meeting existing requirements (Sastry et al. (2023)); many insurers would experience capital shortfalls under adverse climate risk scenarios (Jung et al. (2023)). In response, the NAIC has recently begun developing tools to assess insurers’ climate risk exposures and considering adjustments to capital regulation, including the introduction of insurer climate stress tests.¹⁶ The literature shows that insurers’ risk management crucially depends on whether regulatory rules sufficiently reflect underlying risks: insurers actively hedge only when liabilities recognize risk exposures, but not otherwise, highlighting how regulatory design can directly affect insurers’ risk exposures (Sen (2023)). Regulators have also expanded climate-risk disclosure requirements to promote greater transparency (Liu et al. (2024)). Evaluating these emerging reforms—both their optimal design and their effects on solvency and market structure—remains a ripe area for future research, particularly given concerns that climate change may increasingly strain insurer balance sheets and threaten their solvency.

Rate (price) regulation. In the U.S., regulators have responded to concerns about market power by regulating insurance premiums. Regulation occurs state by state; Oh et al.

¹⁵The fact that insurance is regulated at the state level can create complex coordination challenges because there may be spillovers and externalities between regions (Tiebout (1956)).

¹⁶See the NAIC’s climate scenario analysis and “RCAT Catastrophe Risk Charge.”

(2025) show that states vary starkly in how easily insurers can adjust prices; in a subset of “high friction” states, insurers are unable to fully adjust prices in response to losses. To alleviate these pricing frictions, insurers cross-subsidize by raising prices in “low friction” states, as financial constraints cause insurers to link their pricing decisions across states. These asymmetric adjustments result in premiums to diverge from underlying risks, with households in less regulated states effectively subsidizing those in more regulated states. Oh et al. (2025) also show that rate regulation results in insurer exits and an expansion of insurer of last resort markets. In the same vein, Liu and Liu (2024) show that rate regulation erodes insurers’ financial health and leads to exits in the long term care insurance market.

Another important margin of adjustment is insurers’ asset allocation. For example, Fringuellotti et al. (2024) shows that insurers partly offset pricing frictions by reaching for yield in their investment portfolios. This behavior can be interpreted as another form of cross-subsidization: if riskier portfolio choices lead to insurer distress and failures, the resulting costs are borne by all policyholders, not just those in high friction states.

Several open questions remain, including the microfoundations of rate regulation, its effectiveness in addresses market power, its interactions with other policy and economic frictions, and how rate regulation should be optimally designed given the challenges of climate change. A central question concerns the political economy of rate regulation and how coordination externalities between states can be addressed. For example, Liu and Liu (2024) shows that political frictions materially affect insurance premium increase requests. We also note that there may be interesting connections to the literature on tax competition across states (Agrawal et al. (2025)), with the caveat that the existence of cross-state subsidization may make the insurance setting different.

Residual markets: Property insurance markets have long experienced rationing and exit, well before concerns about climate change became commonplace.¹⁷ As a natural response to diminished availability, a subset of states developed “residual markets”, which are state-run insurance companies of last resort for households that are unable to secure private insurance.¹⁸ Residual insurance markets are sizable in some states, such as Florida, and have expanded rapidly in others, including California. Yet, despite their importance, they remain particularly understudied. Abramson et al. (2025) examine how residual markets price contracts and are funded, showing that the interaction of residual markets and rate regulation influences pricing and insurer exit and how underpriced government policies can

¹⁷For example, private insurers stopped providing flood insurance in the 1960s-1970s; homeowners insurers exited areas subject to heightened riot risk in the 1970s; and they exited the terrorism risk insurance market following the 9/11 attacks.

¹⁸Ellen et al. (2025) provides a history of residual markets in the US, finding that the design of the market unintentionally incentivized owners of multi-family properties to commit arson in the 1970s.

crowd out private insurers. [Taylor et al. \(2025\)](#) find that a temporary moratorium on insurer exit in California had the unintended effect of prompting even more exit and accelerating the growth of the residual market.

One key concern is whether the growth of residual markets poses fiscal stability threats for state and local governments. In Florida, concerns about fiscal exposures through residual markets led the state to adopt a “Depopulation” program, where the state-run insurer transferred policies to private sector entities in order to shrink their balance sheet ([Sastry et al. \(2023\)](#)). Such Depopulation programs are now being implemented in a number of other states, including Louisiana. Looking ahead, it is likely that residual markets will play an increasingly large role in state and local balance sheets if private insurers continue to exit high risk areas. Understanding the financial economics of residual markets, including its broader macroeconomic effects will only grow in importance.

Guaranty funds. Another key feature of U.S. insurance markets is the presence of state guaranty funds, that are designed to protect policyholders in the event that an insurer becomes insolvent. Policymakers view guaranty funds as essential for consumer protection, given the opacity of insurer balance sheets and households’ limited ability to monitor solvency risk. However, evidence shows that guaranty funds create moral hazard, contributing to excessive premium growth prior to insolvency, lower reserves, and riskier asset holdings ([Cummins \(1988\)](#), [Hall and Bohn \(1999\)](#), [Lee and Smith \(1999\)](#), and [Lee et al. \(1997\)](#)). In recent work, [Sastry et al. \(2025\)](#) show that the structure of guaranty funds can lead to a market unraveling where fragile insurers do not fully internalize the costs they impose on other insurers through the fund. This makes it difficult for larger, more diversified insurers to compete on price, and also gives them strong incentives to exit in order to limit their exposure to guaranty fund assessments.

Despite their ubiquity, there is little research on guaranty funds, let alone how they may be impacted by climate change. Questions of particular interest include how effective are these institutions in protecting households; how should they be optimally designed given the growing strain that climate losses may place on insurer balance sheets. Another question is whether guaranty funds are adequately funded, a question particularly relevant as more insurers become financially vulnerable.¹⁹ While there are likely parallels with other explicit and implicit bailout schemes, such as the deposit insurance literature ([Diamond and Dybvig \(1983\)](#), [Dávila and Goldstein \(2023\)](#)), we believe the structure and the economic rationale of guaranty funds make this setting particularly different in its theoretical underpinnings.

Overall, there are a number of common questions that cut across all four policies we

¹⁹[Kojen and Yogo \(2016\)](#) show that the insolvency risk arising from the growth of less capitalized captive re-insurers account for a significant share of the life insurers’ guaranty fund financing pool.

discuss. We believe that understanding the rationale for these policies, the impact of policy interventions on equilibrium outcomes in insurance markets, as well as their interaction with each other are particularly important questions for future research. We welcome work on whether these policies create more costs than benefits, from both a fiscal standpoint as well as in a welfare context.

3.4. Market Incompleteness

The standard model implicitly assumes that financial markets are complete. However, insurance markets feature a number of sources of incompleteness. As a practical matter, it is nearly impossible to lock-in a contract for the long-term; contracts are typically 1-2 years, with no guarantees that the contract will be renewed or that the prices will remain stable. This is in contrast to the mortgage market, where mortgage lenders offer much longer maturity fixed-rate contracts. There is evidence that household demand for such longer-term insurance contracts exists ([Kunreuther and Michel-Kerjan \(2015\)](#)). As a result of this institutional feature, households bear significant volatility in insurance prices, potentially leading to welfare losses. These issues will become more pronounced as climate change is likely to reshape risk classifications: some areas will become more risky while others become less so, with insurance prices adjusting accordingly. However, insurance against such reclassification risk does not exist ([Cochrane \(1995\)](#), [Gruber and Solomon \(2025\)](#)).

Another issue is that there may be concerns about insurers carving out particular perils or rationing coverage amounts. For example, in recent years, private insurers in California and Florida have started to carve out fire and wind damage from standard contracts. Another source of concern is the incompleteness coming from private insurers exiting the market. Although residual markets provide policyholders a safety net when private insurers exit, these contracts typically have limited features. In this vein, after private insurers withdrew from the flood market in the 1970s, the government created a backstop, but coverage limits still leave households exposed to significant uninsurable flood risk ([Sastry \(2025\)](#)).

Understanding the economic drivers for these missing markets is a first order question for the insurance literature. As we discuss earlier, rationing can occur optimally if insurers face a combination of financial constraints, correlated risks, or if they face uncertainty over the loss distribution. Policy frictions, such as rate regulation, can also generate market incompleteness. Classic theories of informational asymmetries can explain market incompleteness and unraveling, a topic we discuss in Section 4.3. Other time-inconsistency failures (e.g. [Cochrane \(1995\)](#)), commitment or coordination challenges, may also be relevant.

Limits to diversification. A related concern is that there may be limits to diversifi-

cation, either through geographic expansion or capital markets, which may lead constrain insurers’ supply choices and affect insurance pricing. Indeed, many insurers are not geographically diversified (Sastry et al. (2023)). Why under-diversified insurers gain large market shares and why competitive pressures do not induce more diversified insurers to enter, remains an open question.²⁰

In theory, insurers could also achieve diversification through reinsurance markets or catastrophe bond markets, which play a critical role in insurance supply by absorbing large correlated losses. However, diversifying risks through reinsurance or catastrophe bonds remains challenging and costly (Froot and O’Connell (2008), Tomunen (2025)), leading to low take-up rates. For example, evidence suggests that catastrophe bond securities are highly illiquid and not strictly zero-beta due to market frictions (Tomunen (2025)).

We believe that understanding the limits to diversification is a first-order question for the property insurance literature. Despite their importance, reinsurance markets remain highly understudied (Froot et al. (1999), Keys and Mulder (2024)). Understanding the factors that influence the resilience and capacity of reinsurance markets under growing climate losses, as well as how reinsurance affects insurance availability, pricing, and solvency remain important open questions. Future work should also examine how catastrophe bonds are priced, who holds them, and how they influence the pricing of insurance contracts.

4. Demand for Insurance

In this section, we consider a number of deviations from the benchmark model that influence household demand for insurance. Section 4.1 focuses on a central feature of the homeowners insurance market, that insurance demand is determined jointly by households and mortgage lenders, because most households purchase their homes with a mortgage. We also consider the impact of financing frictions (Section 4.2) and informational and behavioral frictions (Section 4.3), which also influence household and lender insurance demand. We survey the literature on each deviation, where available, and provide speculative thoughts into how climate change may shape the demand-side of the market.

4.1. Mortgage Market Effects

A unique feature of property insurance markets is that they are intimately connected with mortgage markets. The average borrower puts roughly 20% down on their mortgage at orig-

²⁰One rationale for insurers having geographically concentrated portfolios could be returns from specialization. Related work on banks indicates that they frequently specialize in certain industries, leveraging comparative advantage and expertise (Duquerroy et al. (2022), Blickle et al. (2021), Blickle et al. (2025)).

ination, while lenders or mortgage owners retain the other 80%. Therefore, mortgage lenders can be exposed to significant losses if borrowers default after natural disasters and the properties collateralizing the mortgage are also damaged. Insurance is important for protecting lender exposures in natural disasters. [Sastry et al. \(2023\)](#) shows that fragile insurers—those at high risk of insolvency—cause higher mortgage defaults after natural disasters than financially stable insurers, explaining why lenders have incentives to screen for insurance quality.

Because lenders have an interest in preserving the value of the collateral, mortgage lenders ubiquitously require mortgage borrowers to purchase homeowners insurance as a condition of the loan. They are also listed as beneficiaries on the insurance contract.²¹ These requirements lead to very high take-up rates in the property insurance markets, with nearly 83% of households having homeowners insurance ([Sastry et al. \(2024\)](#)).

Beyond simply requiring insurance, mortgage lenders often screen which insurance companies they are willing to accept, influencing the industrial organization of the homeowners insurance market. [Sastry et al. \(2023\)](#) find a stark market segmentation, where the market share of fragile insurers is high for GSE-eligible conforming mortgages but low for non-conforming mortgages retained by private lenders. They argue this segmentation emerges because private lenders have more stringent screening of insurance insolvency risk than the GSEs (Fannie Mae and Freddie Mac).

Lenders also require households to buy enough insurance coverage to protect their loan exposures; as a result, a large component of the demand for insurance actually comes from mortgage lenders. [Sastry et al. \(2024\)](#) show that lender minimum coverage requirements play an important role in household coverage choices. Furthermore, they find that households use mortgage credit to finance insurance coverage beyond lender minimums. These results contrast with other well-known settings, such as health or flood insurance. While it is true that households with debt have reduced incentives to buy insurance when they benefit from the implicit default option ([Mahoney \(2015\)](#), [Liao and Mulder \(2021\)](#), [Bakkensen et al. \(2025\)](#), [Bailey et al. \(2019\)](#)), in the homeowners insurance context lenders endogenously respond to these incentives through strictly enforced insurance requirements.

Given the out-sized importance of mortgage lenders for the insurance market, the topic remains particularly understudied. In particular, there are questions about optimal insurance and mortgage contract design in light of these complex institutional arrangements. There may be relevant parallels with the development-economics literature on crop insurance, which studies optimal bundling of insurance and credit under financial and informational frictions (e.g., [Braverman and Stiglitz \(1982\)](#); [Banerjee et al. \(2014\)](#)). Another question is how mort-

²¹Households also use insurance proceeds to repair homes after disasters which improves collateral values, benefiting lenders ([Issler et al. \(2020\)](#)).

gage and bank regulation should account for frictions in insurance markets when quantifying climate risk in the financial system. We also note that there are interesting parallels to the health insurance literature, which examines how hospital-insurer interactions shape outcomes in both health care and insurance markets (Ho and Lee (2017)).

4.2. Financial Frictions

Many intuitions of the standard model change when households are assumed to be credit constrained, although the predictions on *how* credit constraints should influence demand are not obvious. In one class of models, financial constraints *increase* demand for insurance, because insurance becomes more valuable when households cannot rely on credit markets to smooth shocks (Finkelstein et al. (2019)). On the other hand, financial constraints can also *reduce* demand for insurance, because households are unable to finance insurance purchases using credit (Rampini and Viswanathan (2013), Ericson and Sydnor (2018), Casaburi and Willis (2018), Orestes et al. (2025)). For financially constrained households, which channel dominates ultimately becomes an empirical question, and can change depending on how financial constraints interact with rising climate risks and the types of insurance and credit products households use to smooth shocks. In this vein, Sastry et al. (2024) argues that demand elasticities in homeowners insurance are a function of households' debt capacity. We expect future research to shed light on how climate risk may exacerbate household financial constraints and affect the demand for property insurance.

4.3. Informational and Behavioral Frictions

Insurance and credit markets can exhibit rationing and market failures when households hold private information about their own risk types or actions (Akerlof (1970); Rothschild and Stiglitz (1976); Stiglitz and Weiss (1981); Bester (1985)). A rich literature examines the presence and forms of private information in insurance markets.²² Even when risks are observable, constraints on risk-based pricing can produce outcomes that resemble adverse selection (Finkelstein and Poterba (2014a)). Likewise, policy restrictions can create ex-ante moral hazard (Ehrlich and Becker (1972)), encouraging households to locate in higher-risk areas when insurance is mispriced. Ex-post moral hazard can occur if households with insurance contracts are less likely to invest in maintenance or resiliency (Pauly (1968)).

Private Information. There is limited consensus on the extent of private information

²²An extensive literature examines the topic of selection and private information, including testing for private information and quantifying its welfare costs (Einav and Finkelstein (2011), Einav et al. (2012), Chetty and Finkelstein (2013), Finkelstein and Poterba (2014b), Chiappori and Salanie (2000), Finkelstein and Poterba (2014a), Guerrieri and Shimer (2014)).

in homeowners insurance and whether it leads to market failures. On the one hand, the effects of private information should be minimal because observable features such as location and structure characteristics are the key drivers of losses (Jaffee and Russell (1997)). On the other hand, households may react to the same disaster shock differently depending on unobservable characteristics (e.g., risk aversion, financial constraints) (Blonz et al. (2024), Cookson et al. (2023)). The information advantage may even be reversed: insurers instead may hold an informational advantage due to proprietary catastrophe models and granular loss data (Brunnermeier et al. (2021); Boomhower et al. (2023)). There is emerging evidence on private information and the role of pricing restrictions, mostly from the flood insurance context (Wagner (2022), Bradt et al. (2021)), future work is needed to clarify the nature of selection in homeowners insurance.

A key institutional feature is that mortgage lenders require borrowers to maintain homeowners insurance, effectively creating an *individual mandate* that limits selection on the extensive margin.²³ By contrast, take-up is discretionary for households who purchase their homes without a mortgage, and thus may be subject to selection. Moreover, take up rates could change in the future if lenders choose to selectively enforce the insurance requirement as climate losses place burden on household budgets, which may alter the nature of selection even among mortgage borrowers.²⁴ Future work could help clarify the severity of private information and the relative roles of observable versus unobservable determinants of risk.

Moral Hazard from Disaster Aid. Large disasters trigger substantial explicit and implicit transfers through grants, subsidized loans, tax relief (Gallagher and Hartley (2017); Billings et al. (2019); Collier et al. (2024)), social insurance programs (Deryugina (2017)), or mortgage forbearance or loan modifications (Panjwani (2024)). Households that expect to receive such transfers may have lower demand for private insurance (Michel-Kerjan (2010)).²⁵ However, there is limited empirical evidence on whether the moral hazard from public aid crowds out insurance demand. There is some evidence that households do not expect to receive much aid (Kunreuther et al. (1978)), while Kousky et al. (2018) show that disaster assistance reduces flood coverage on the intensive margin. In practice, disaster assistance is typically modest and tightly screened (Kousky (2013), Wagner (2022)), which imply limited crowd out. Understanding how disaster aid affects insurance demand remains an open question with implications for the optimal design of disaster aid policies (Aron-Dine (2025)).

²³The mandate applies to all borrowers, not only those in high-risk areas, and evidence shows it is highly enforced. For example, nearly all mortgage borrowers hold homeowners insurance (Sastry et al. (2025)).

²⁴For example, there is some concern that mortgage lenders selectively enforce the requirement that borrowers in FEMA flood zones buy flood insurance (FEMA, 2018).

²⁵Relatedly, see Finkelstein et al. (2019) and Finkelstein et al. (2018) for discussion of how uncompensated care or subsidized public insurance programs can crowd out private health insurance purchases.

Beliefs and Behavioral Frictions. Household beliefs and risk preferences also influence their willingness-to-pay. As climate losses rise, households may become more risk averse. For instance, [Barseghyan et al. \(2011\)](#) show that households exhibit greater risk aversion in homeowners than in auto insurance. Evidence suggests that households systematically underestimate their own risk exposures (e.g., for flood risk [Bakkensen and Barrage \(2022\)](#), [Wagner \(2022\)](#)). While it is well known that insurers rely on proprietary data and catastrophe models, how households form beliefs is less well understood. Evidence suggests that several factors likely shape household beliefs: disclosure mandates ([Doerner et al. \(2025\)](#), [Bakkensen and Ma \(2020\)](#)), information provided during the housing search process ([Fairweather et al. \(2024\)](#)), government risk maps ([Mulder \(2022\)](#)), or insurance prices ([Ge et al. \(2023\)](#)). Political views may also influence household beliefs about risk, consistent with evidence on partisan sorting in the housing market ([Baldauf et al. \(2020\)](#), [Bernstein et al. \(2020\)](#)). Understanding how households form beliefs, and the role of information disclosure in that process, remains an important direction for future research.

Behavioral frictions could also shape household demand for financial products, including insurance ([Beshears et al. \(2018\)](#)). For example, households often neglect tail risks, lowering willingness-to-pay ([Kunreuther and Pauly \(2004\)](#), [Botzen et al. \(2015\)](#), [Bakkensen and Barrage \(2022\)](#)). Attention also affects demand. For example, households increase flood insurance coverage when disasters are salient ([Gallagher \(2014\)](#), [Hu \(2022\)](#), [Giglio et al. \(2021\)](#), [Dessaint and Matray \(2017\)](#)). Limited financial sophistication can further influence demand ([Collier et al. \(2024\)](#)).²⁶ [Cookson et al. \(2025\)](#) find that households buy lower coverage as they compare insurers based on monthly premiums, rather than the premium per dollar of coverage. Given the complexity of insurance contracts, more work is needed on how behavioral frictions and financial sophistication shape demand.

Insurance demand depends on trust that insurers will actually pay claims. Yet evidence points to imperfect trust, especially in emerging markets where weak contract enforcement is a central concern ([Karlan et al. \(2014\)](#)). Even in the U.S., valid homeowners insurance claims are often disputed or denied ([Gennaioli et al. \(2021\)](#)), and insurers strategically delay payouts to manage post-disaster capital constraints ([Jotikasthira et al. \(2025\)](#)). Such concerns help explain low willingness-to-pay in other markets ([Briggs et al. \(2023\)](#)). Perceptions of insurer insolvency also shape household trust ([Tennyson and Inman \(1997\)](#)). Future research can examine the drivers of trust, and how it may evolve as climate change places increasing strain

²⁶For example, many homeowners mistakenly believe that their standard homeowners insurance policy covers floods, even though flood damage is typically carved out. [The Insurance Journal](#), “[Survey: Most Homeowners Believe Their Policy Covers Flood](#)”, June 20, 2024. Similarly, the National Association of Insurance Commissioners (NAIC) reported that 56% of U.S. households believe that flood damage is covered by standard homeowners insurance.

on insurer balance sheets.

5. Insurance and the Macroeconomy

Property insurance influences the macroeconomy through a number of channels, some of which are just beginning to be explored in the literature. In this section, we outline a few channels based on existing literature, but note that there may be many additional channels that future research may identify.

Housing markets. Insurance plays an important role in the housing market. As we discussed in Section 4.1, insurance is required to obtain a mortgage, and most households rely on mortgages to purchase homes. A rich literature studies how credit supply impacts housing markets (e.g., Favara and Imbs (2015)), but the literature on the role of property insurance is only beginning to emerge. There is significant evidence that higher flood insurance and homeowners insurance premiums lead to house price declines, even conditioning on risk (Kousky et al. (2020), Bakkensen and Barrage (2022), Hino and Burke (2021), Georgic and Klaiber (2022), Ge et al. (2023), Keys and Mulder (2024), Eastman et al. (2024)). Theoretically speaking, there are a number of potential channels, including that insurance premiums are a fundamental component of the user cost of housing, and that households may learn about risk from insurance premiums. Insurance availability also influences credit supply (Sastry (2025), Hu (2024)), which may assert independent effects on housing demand and prices. Insurance can also change housing construction, including the quantity and quality of homes which are built, and where they are built (Ostriker and Russo (2022), Issler et al. (2020), Fabian (2024)). There is some recent evidence that large well-diversified institutional investors have begun to step in and buy multi-family properties in high climate risk areas (CNBC (2025)), changing the ownership composition of housing. Relatedly, private equity firms are buying insurance companies, including P&C subsidiaries, with Blackstone even issuing its own catastrophe bond (Swiss Re (2023)). We believe that the link between housing and insurance markets is a particularly promising line of research.

Mispricing and economic activity. As we discuss in Section 2, there is an important relationship between insurance prices and the geography of economic activity. In a frictionless setting where insurance is priced accurately, we argue that a first-best allocation would be achieved. Yet, much of the literature notes that in practice, insurance is likely to be heavily mis-priced, with premiums deviating from actuarially fair rates. For example, it is well-known that government-provided flood insurance is deliberately subsidized (Wagner (2022)), which leads to significant sorting of lower income households into risky areas (Bakkensen and Ma (2020)). Additionally, homeowners insurance is frequently subject to rate regulation and

other policy-related frictions that impact pricing and availability (Oh et al. (2025), Sastry et al. (2025), Abramson et al. (2025)).

Such pricing distortions can have pervasive effects on real economic decisions by households and firms. Under-priced insurance can incentivize living in high risk areas, creating a type of ex-ante moral hazard (Ehrlich and Becker (1972)). In this vein, Sastry et al. (2023) find that insurance mispricing and agency problems in mortgages impact mortgage origination in high risk areas. Additionally, Peralta and Scott (2024), Ge et al. (2024) and Fabian (2024) finds that insurance premiums have a direct impact on household location choice and local economic development. Fried (2021), Bilal and Rossi-Hansberg (2021), and Bilal and Rossi-Hansberg (2023) show that households can rely on real investments and migration to manage climate risks when households are financially constrained or financial markets are incomplete, leading to large re-allocation across regions as climate risks intensify.

Interestingly, in some contexts, inefficiencies in insurance markets can be somewhat mitigated in mortgage markets. For example, Sastry (2025) find that lenders require higher down payments for borrowers in flood zones with uninsurable flood risk; these requirements change the distribution of mortgage borrowers in flood zones towards richer and higher credit quality individuals, offsetting some of the distortions stemming from subsidized flood insurance. Similarly, Sastry et al. (2023) find that private mortgage lenders are aware when fragile insurers under-price risk, and that they account for the enhanced insolvency risk in mortgage pricing and credit supply.

While mis-pricing can create distortions, moving toward an economy where insurance pricing better reflects true underlying risks also presents several challenges. A first issue is that under-insurance remains prevalent. Sastry et al. (2024) finds that financially constrained households react to rising homeowners insurance premiums by reducing coverage, regardless of whether prices are actuarially fair or not. There is similar evidence from flood insurance, where moves toward risk-based pricing led households to drop insurance coverage altogether (Wagner (2022), Mulder and Kousky (2023), Gruber and Solomon (2025)). A related concern is that rising premiums can trigger sharp adjustments in mortgage availability and lead to large and quick declines in house prices. Furthermore, rising insurance prices can worsen mortgage risks in a number of ways, including incentivizing mortgage default and prepayment (Gallagher and Hartley (2017), Biswas et al. (2023), Ge et al. (2024)). These dynamics would make it more difficult for households in high risk areas to move away, a key consideration in the debate between investing in resiliency and encouraging managed retreat. We expect future research to shed light on the complex ways that insurance and mortgage contracts interact to influence household mobility and location choice, and how this may lead to new institutional arrangements in the housing market.

Fiscal and Financial Stability. Mis-pricing in insurance markets can also create fiscal risks as well as financial stability risks. [Sastry et al. \(2023\)](#) find that Fannie Mae and Freddie Mac, which own or guarantee nearly two-thirds of the \$15 trillion mortgage market, have significant exposures to fragile insurers that are at high risk of insolvency, creating fiscal risks for taxpayers because insurer insolvencies cause heightened mortgage default after disasters. Furthermore, the GSEs are systemically important institutions, meaning that extreme losses can require taxpayer bailouts. Furthermore, the growth of state-run insurers-of-last-resort pose another threat to fiscal stability of state and local governments, especially if they under-price risks relative to actuarially fair rates.

Climate adaptation. From a theoretical standpoint, insurance companies should consider household or municipality investments in resiliency when forming their risk assessments and determining prices. However, regulatory constraints on pricing may prevent them from doing so, which in turn may reduce incentives to invest in climate adaptation infrastructure. However, even under such pricing regulations, there may be incentives for insurers with large concentrated exposures to particular geographies to invest in public goods like adaptive infrastructure, in the spirit of [Kojen and Van Nieuwerburgh \(2020\)](#) who discuss how life insurance companies have incentives to subsidize R&D for life-extending drugs.

Geographic transmission of shocks. Insurance enables interregional risk sharing through the pooling of local idiosyncratic risks by well-diversified insurers, helping to create potential welfare gains. On the other hand, interregional connections can lead shocks in one region to spill over connected areas through the insurance network ([Oh et al., 2025](#)). An added layer of complexity comes from [Gupta et al. \(2025\)](#), who show that when insurance markets are local but consumption markets are integrated, the standard intuitions about insurance demand become complicated, and that collapses of insurance in one region can spread to other markets. Similarly, [Sastry et al. \(2025\)](#) show that insurance insolvencies in one state can lead to changes of insurance supply in other states because of externalities through guaranty funds. We expect future research to further explore the economics of geographic spillovers through property insurance.²⁷

6. Additional Topics and Conclusion

Our review has focused primarily on U.S. homeowners insurance. We recognize, however, that the challenge of insuring climate risk is far broader than we could cover here. Below,

²⁷These questions have been relatively well studied in banking and capital markets ([Allen and Gale \(2000\)](#), [Asdrubali et al. \(1996\)](#), [Hurst et al. \(2016\)](#), [Landier et al. \(2017\)](#), [Cortés and Strahan \(2017\)](#), [Aguirregabiria et al. \(2025\)](#)), however, evidence from insurance markets is scarce.

we highlight additional questions, which are also important but beyond our current scope.

Commercial real estate. Commercial insurance markets, including the commercial real estate sector, have explored new risk-transfer tools to manage rising climate losses. Understanding the exposures of firms and the extent to which commercial insurance markets help mitigate exposures is largely unexplored.

Sovereign risk. Another question concerns designing products for sovereigns facing concentrated climate exposure, such as small island states, or emerging markets that face financial constraints. While there is work on crop insurance in these settings, there is less work on understanding the most commonly used insurance products for governments – including blended financed mechanisms, sovereign risk pools, and parametric insurance products.

Asset allocation. Finally, although our focus has been on insurers’ liabilities, insurance companies also hold large investment portfolios. Their asset allocation may be shaped by liability-side climate exposures, yet research on property insurers’ investment behavior remains limited compared to the life insurance literature. We expect future work to consider how the asset side of insurers’ balance sheets may connect with their liabilities exposures and how climate risk may influence asset allocation decisions.

Data advances and conclusion. Recent data improvements have transformed the study of property insurance. Just a few years ago, the best available homeowners insurance data consisted of insurer–state–year aggregates from the NAIC, and NFIP flood insurance data was not systematically shared with researchers. Today, commercial providers such as ICE McDash offer detailed datasets that can be linked to mortgage and borrower characteristics. The Federal Insurance Office-NAIC provide national ZIP-level claims information, and Quadrant Information Services provide ZIP–insurer–level pricing data. Similarly, flood insurance premiums and claims are accessible through a public API, and several states now release sub-state granular homeowners insurance data.

It has never been a better time to study property insurance markets. The data advances now allow empirical evaluation of questions that was previously not possible. At the same time, recent events have called us to revisit our understanding of the economics of property insurance, and provide policy solutions to address critical challenges facing these markets. The fast pace at which physical risks have intensified makes answering these questions particularly urgent. We hope our review can be a useful tool for researchers seeking to explore these issues, and we look forward to seeing how the literature develops.

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